

Assignment 3

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```
#Load packages
#install.packages("ggplot2")
library(ggplot2)
```

Question 1: Moths

(a) What are the logits in this study?

- The logits are just the log of the odds that a moth gets removed by a predator. So, $\log(p \text{ removed} / p \text{ not removed})$. In this case, we're looking at both light and dark morphs and how likely they are to get removed depending on how far they are from Liverpool.

(b) Create a model with DISTANCE and dark. Interpret all the coefficients.

```
# Load necessary packages
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v lubridate  1.9.4      v tibble    3.3.0
## v purrr      1.0.4      v tidyr     1.3.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
# Read the data
data <- read_csv("moth.csv")
```

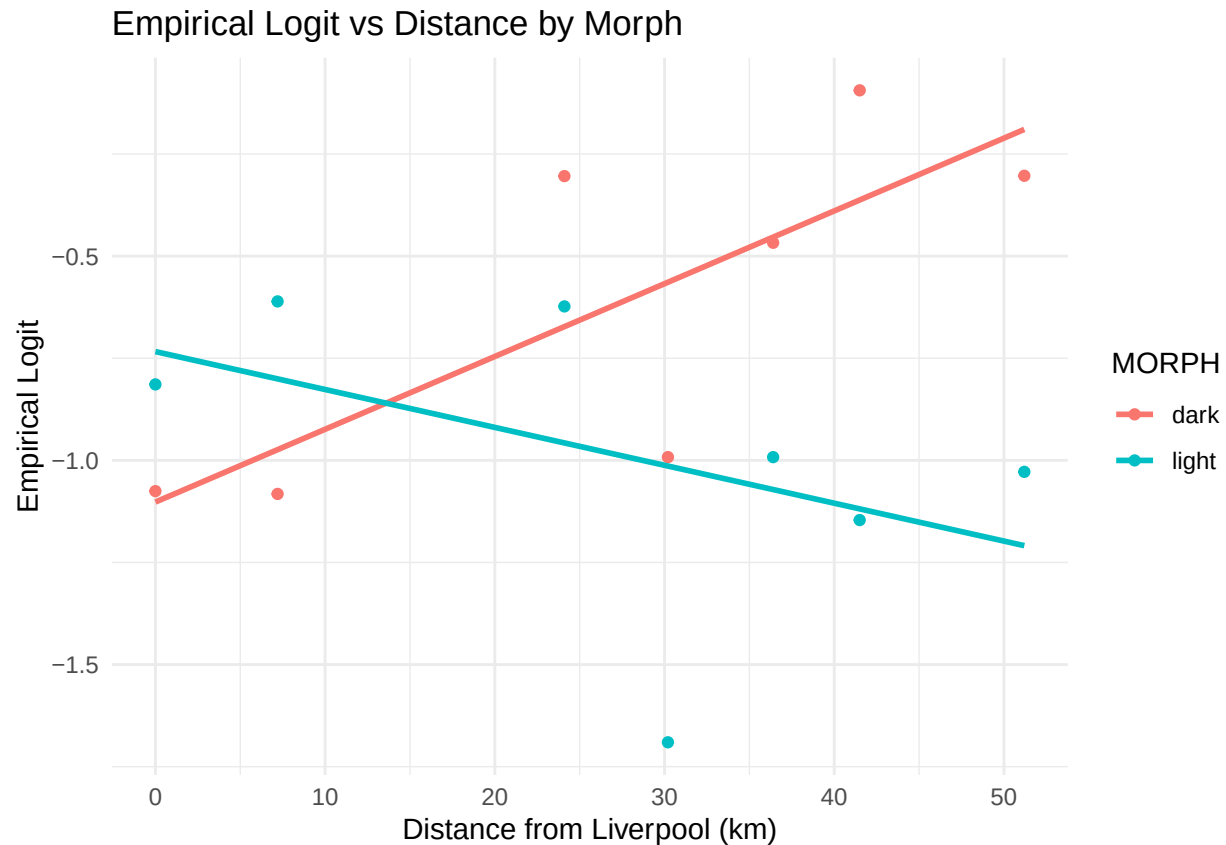
```
## Rows: 14 Columns: 4
## -- Column specification -----
## Delimiter: ","
## chr (1): MORPH
## dbl (3): DISTANCE, PLACED, REMOVED
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
# Peek at the data
head(data)
```

```
## # A tibble: 6 x 4
##   MORPH DISTANCE PLACED REMOVED
##   <chr>    <dbl>  <dbl>  <dbl>
## 1 light      0      56     17
## 2 dark      0      56     14
## 3 light    7.2     80     28
## 4 dark    7.2     80     20
## 5 light   24.1     52     18
## 6 dark   24.1     52     22
```

```
# Calculate empirical logits for each morph at each location
data <- data %>%
  mutate(
    logit = log((REMOVED + 0.5) / (PLACED - REMOVED + 0.5))
  )
# Plot empirical logits vs distance, by morph
ggplot(data, aes(x = DISTANCE, y = logit, color = MORPH)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Empirical Logit vs Distance by Morph",
    x = "Distance from Liverpool (km)",
    y = "Empirical Logit"
  ) +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



- From the plot, it looks like the light morphs are more likely to be removed near Liverpool since their empirical logits start higher and decrease as the distance increases. For dark morphs, the trend is flatter. This supports the idea that light moths stand out more near Liverpool, where tree trunks were dark from pollution, so predators spotted and removed them more easily.

(c)) Create a model with **DISTANCE**, **dark**, and the interaction between both variables. Interpret all the coefficients.

```
# Ensure MORPH is a factor
data$MORPH <- factor(data$MORPH)
# Logistic regression model
model <- glm(
  cbind(REMOVED, PLACED - REMOVED) ~ DISTANCE + MORPH,
  data = data,
  family = binomial
)
# Output model summary
summary(model)
```

```
##
## Call:
## glm(formula = cbind(REMOVED, PLACED - REMOVED) ~ DISTANCE + MORPH,
##      family = binomial, data = data)
```

```
##
## Coefficients:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.732690   0.151221  -4.845 1.27e-06 ***
## DISTANCE     0.005314   0.004002   1.328 0.18422
## MORPHlight  -0.404052   0.139377  -2.899 0.00374 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 35.385  on 13  degrees of freedom
## Residual deviance: 25.161  on 11  degrees of freedom
## AIC: 93.836
##
## Number of Fisher Scoring iterations: 4
```

- The intercept is -0.733, which shows the log odds of a dark moth being removed right at Liverpool (distance = 0).
- The estimate for distance is 0.0053, but the p-value is 0.184. Since this is greater than the usual alpha level of 0.05, the distance is not statistically significant. So, distance doesn't have a strong effect on removal odds when moth type stays the same.
- For the light moths, the estimate is -0.404 with a p-value of 0.0037. Since this is less than 0.05, it's statistically significant. That means light moths are less likely to be removed compared to dark ones, no matter how far the location is from Liverpool.

(d) Create a model with DISTANCE, dark, and the interaction between both variables. Interpret all the coefficients.

```
# Model with interaction between DISTANCE and MORPH
model_d <- glm(
  cbind(REMOVED, PLACED - REMOVED) ~ DISTANCE * MORPH,
  data = data,
  family = binomial
)
# Display summary
summary(model_d)
```

```
##
## Call:
## glm(formula = cbind(REMOVED, PLACED - REMOVED) ~ DISTANCE * MORPH,
##      family = binomial, data = data)
##
## Coefficients:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -1.128987   0.197906  -5.705 1.17e-08 ***
## DISTANCE         0.018502   0.005645   3.277 0.001048 **
## MORPHlight       0.411257   0.274490   1.498 0.134066
## DISTANCE:MORPHlight -0.027789  0.008085  -3.437 0.000588 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 35.385  on 13  degrees of freedom
## Residual deviance: 13.230  on 10  degrees of freedom
## AIC: 83.904
##
## Number of Fisher Scoring iterations: 4
```

- This model includes DISTANCE, morph type, and their interaction. Assuming an alpha level of 0.05, we compare all p-values to this threshold to check for significance.
- The intercept is negative 1.13, which represents the log odds of a dark morph moth being removed when the distance is 0 km from Liverpool. It is significant because its p-value is far below 0.05.
- The estimate for DISTANCE is 0.0185, and since its p-value (0.001) is below 0.05, therefore distance has a significant effect. For dark morphs, the further you move from Liverpool, the higher the odds of being removed.
- The coefficient for MORPHlight is 0.41, but its p-value (0.134) is above 0.05, meaning there is no strong evidence of a difference between light and dark morphs at 0 km.
- The interaction term DISTANCE:MORPHlight is negative 0.028 and is significant ($p = 0.0006$). This means the effect of distance on removal odds is weaker for light morphs compared to dark morphs.

(e) Interpret a drop-in-deviance test and a Wald test to test the significance of the interaction term in (d).

```
# Refit model without the interaction term (reduced model)
model_no_interaction <- glm(
  cbind(REMOVED, PLACED - REMOVED) ~ DISTANCE + MORPH,
  data = data,
  family = binomial
)
# Full model from (d), with interaction
model_interaction <- glm(
  cbind(REMOVED, PLACED - REMOVED) ~ DISTANCE * MORPH,
  data = data,
  family = binomial
)
# (1) Drop-in-deviance test: Compare full vs reduced model
anova(model_no_interaction, model_interaction, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: cbind(REMOVED, PLACED - REMOVED) ~ DISTANCE + MORPH
## Model 2: cbind(REMOVED, PLACED - REMOVED) ~ DISTANCE * MORPH
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         11      25.161
## 2         10      13.230  1   11.931 0.0005519 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# (2) Wald test is already in the output from summary(model_interaction)
summary(model_interaction)
```

```
##
## Call:
## glm(formula = cbind(REMOVED, PLACED - REMOVED) ~ DISTANCE * MORPH,
##      family = binomial, data = data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -1.128987    0.197906  -5.705 1.17e-08 ***
## DISTANCE         0.018502    0.005645   3.277 0.001048 **
## MORPHlight       0.411257    0.274490   1.498 0.134066
## DISTANCE:MORPHlight -0.027789    0.008085  -3.437 0.000588 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 35.385  on 13  degrees of freedom
## Residual deviance: 13.230  on 10  degrees of freedom
## AIC: 83.904
##
## Number of Fisher Scoring iterations: 4
```

- Assuming an alpha level of 0.05 and comparing all p-values to that cutoff.
- For the drop-in-deviance test, we compared a reduced model (with only DISTANCE and MORPH) to a full model (with DISTANCE, MORPH, and their interaction). The p-value from the test is 0.0006, which is way below 0.05. This means adding the interaction term significantly improves the model.
- The Wald test, which looks at the p-value from the model summary for the interaction term, also gives a p-value of 0.0006. Since it's less than 0.05, we can again conclude that the interaction between distance and morph is significant. So we should keep the interaction term in the model.

(f) Test the goodness-of-fit for the interaction model. What can we conclude about this model?

```
pchisq(13.230, df = 10, lower.tail = FALSE)
```

```
## [1] 0.2110951
```

- Based on the output, we can conclude that the interaction model fits the data well. The p-value from the Chi-squared goodness-of-fit test is 0.211, which is greater than the assumed alpha level of 0.05. This means we do not reject the null hypothesis that the model fits. So, there's no strong evidence of a lack of fit, and the model appears to be appropriate for the data.

(g) Is there evidence of overdispersion in the interaction model? What factors might lead to overdispersion in this case? Regardless of your answer, repeat (d) adjusting for overdispersion.

```
# calculating the dispersion parameter
dispersion <- 13.230 / 10
dispersion
```

```
## [1] 1.323
```

```
# Refit interaction model adjusting for overdispersion
model_d_over <- glm(
  cbind(REMOVED, PLACED - REMOVED) ~ DISTANCE * MORPH,
  data = data,
  family = quasibinomial
)
# View summary
summary(model_d_over)
```

```
##
## Call:
## glm(formula = cbind(REMOVED, PLACED - REMOVED) ~ DISTANCE * MORPH,
##      family = quasibinomial, data = data)
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.128987    0.223104  -5.060 0.000492 ***
## DISTANCE         0.018502    0.006364   2.907 0.015637 *
## MORPHlight      0.411257    0.309439   1.329 0.213360
## DISTANCE:MORPHlight -0.027789    0.009115  -3.049 0.012278 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasibinomial family taken to be 1.270859)
##
## Null deviance: 35.385  on 13  degrees of freedom
## Residual deviance: 13.230  on 10  degrees of freedom
## AIC: NA
##
## Number of Fisher Scoring iterations: 4
```

- There is no strong evidence of overdispersion in the interaction model. I checked this by dividing the residual deviance by the residual degrees of freedom, which gave a dispersion value of 1.323. Since this is only slightly greater than 1, so its not that bad.
- Overdispersion can happen when there's extra variability in the data that the model doesn't explain. That could be due to missing predictors, clustering in the data, outliers, or other sources of variation not captured by the model.

(h) Compare confidence intervals for coefficients in your models from (g) and (d).

```
# Confidence intervals for model from (d)
confint.default(model_d)
```

```
##                2.5 %      97.5 %
## (Intercept)    -1.516874963 -0.74109814
## DISTANCE       0.007437606  0.02956692
## MORPHlight     -0.126733029  0.94924730
## DISTANCE:MORPHlight -0.043636294 -0.01194179
```

```
# Confidence intervals for overdispersion-adjusted model from (g)
confint.default(model_d_over)
```

```
##                2.5 %      97.5 %
## (Intercept)    -1.566262401 -0.691710702
## DISTANCE       0.006028812  0.030975711
## MORPHlight     -0.195231996  1.017746267
## DISTANCE:MORPHlight -0.045654026 -0.009924062
```

- The actual coefficient estimates stayed the same, but the confidence intervals got a bit wider after adjusting for overdispersion.
- For example, in part d, the confidence interval for the interaction term (DISTANCE:MORPHlight) was from about -0.044 to -0.012, while in the adjusted model, it widened to about -0.0466 to -0.0099. The same pattern shows up for the other terms too.
- This makes sense because when we adjust for overdispersion, we're accounting for extra variability in the data, which increases the standard errors and makes us less confident in the exact size of the effects. But overall, the direction and meaning of the coefficients didn't change.

Question 2: Elephant mating

(a) Create a histogram of MATINGS. Is there preliminary evidence that the number of matings could be modeled as a Poisson response? Explain.

```
# Load data
data <- read.csv("elephant.csv")
# Preview structure and first few rows
str(data)
```

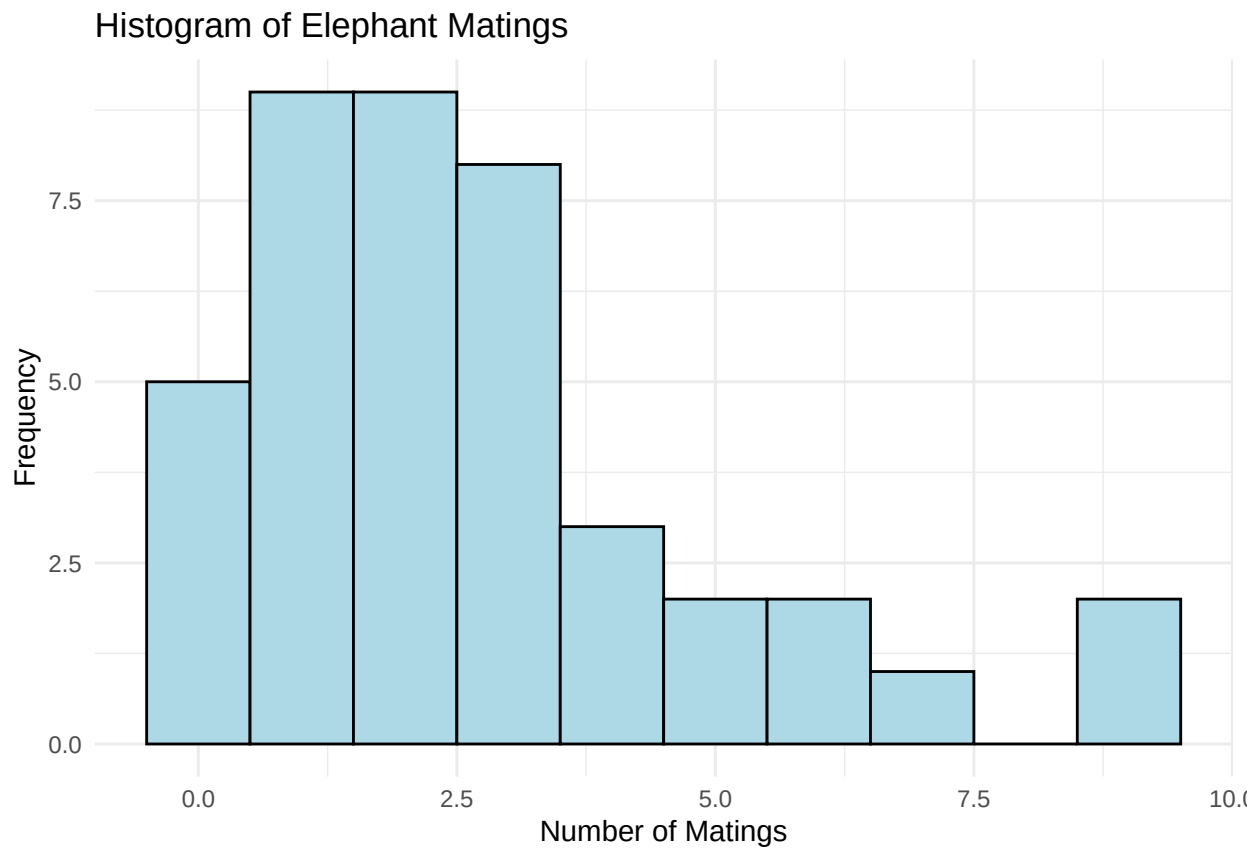
```
## 'data.frame':   41 obs. of  2 variables:
## $ AGE      : num  27 28 28 28 28 29 29 29 29 29 ...
## $ MATINGS: num  0 1 1 1 3 0 0 0 2 2 ...
```

```
head(data)
```



```
## AGE MATINGS
## 1 27 0
## 2 28 1
## 3 28 1
## 4 28 1
## 5 28 3
## 6 29 0
```

```
# Histogram of MATINGS
ggplot(data, aes(x = MATINGS)) +
  geom_histogram(binwidth = 1, color = "black", fill = "lightblue") +
  labs(
    title = "Histogram of Elephant Matings",
    x = "Number of Matings",
    y = "Frequency"
  ) +
  theme_minimal()
```

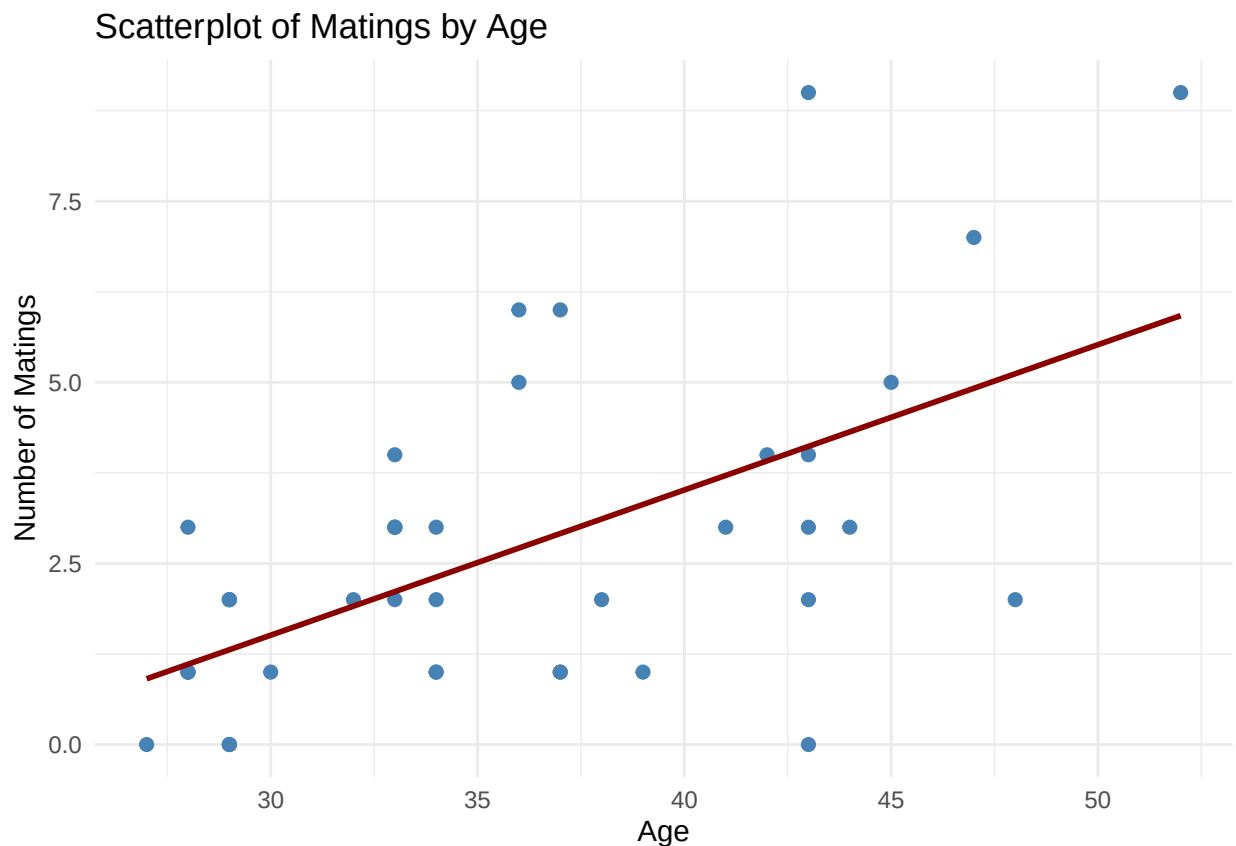


- The histogram of the number of matings shows that the data are right-skewed, with most elephants having a low number of matings (e.g., 0–3), and fewer elephants having higher counts. This pattern is typical of a Poisson distribution, which models count data where the probability of larger values decreases rapidly.
- There are no negative values, and the data appear to be discrete and non-negative integers, which fits the basic requirements of a Poisson model. We can't confirm the mean equals the variance from this graph alone, the shape supports the idea that a Poisson model might be a good starting point for modeling the number of matings.

(b) Plot MATINGS by AGE. Add a least squares line. Is there evidence that modeling matings using a linear regression with age might not be appropriate? Explain.

```
# Scatterplot of MATINGS by AGE with least squares line
ggplot(data, aes(x = AGE, y = MATINGS)) +
  geom_point(color = "steelblue", size = 2) +
  geom_smooth(method = "lm", se = FALSE, color = "darkred") +
  labs(
    title = "Scatterplot of Matings by Age",
    x      = "Age",
    y      = "Number of Matings"
  ) +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



- There's a positive trend, so age does seem to affect the number of matings. But the points are spread out and the data is discrete. There's also a lot of variation that isn't explained by the line. Since the response is a count variable, a linear model might not be the best fit. A Poisson or another count model would make more sense here.

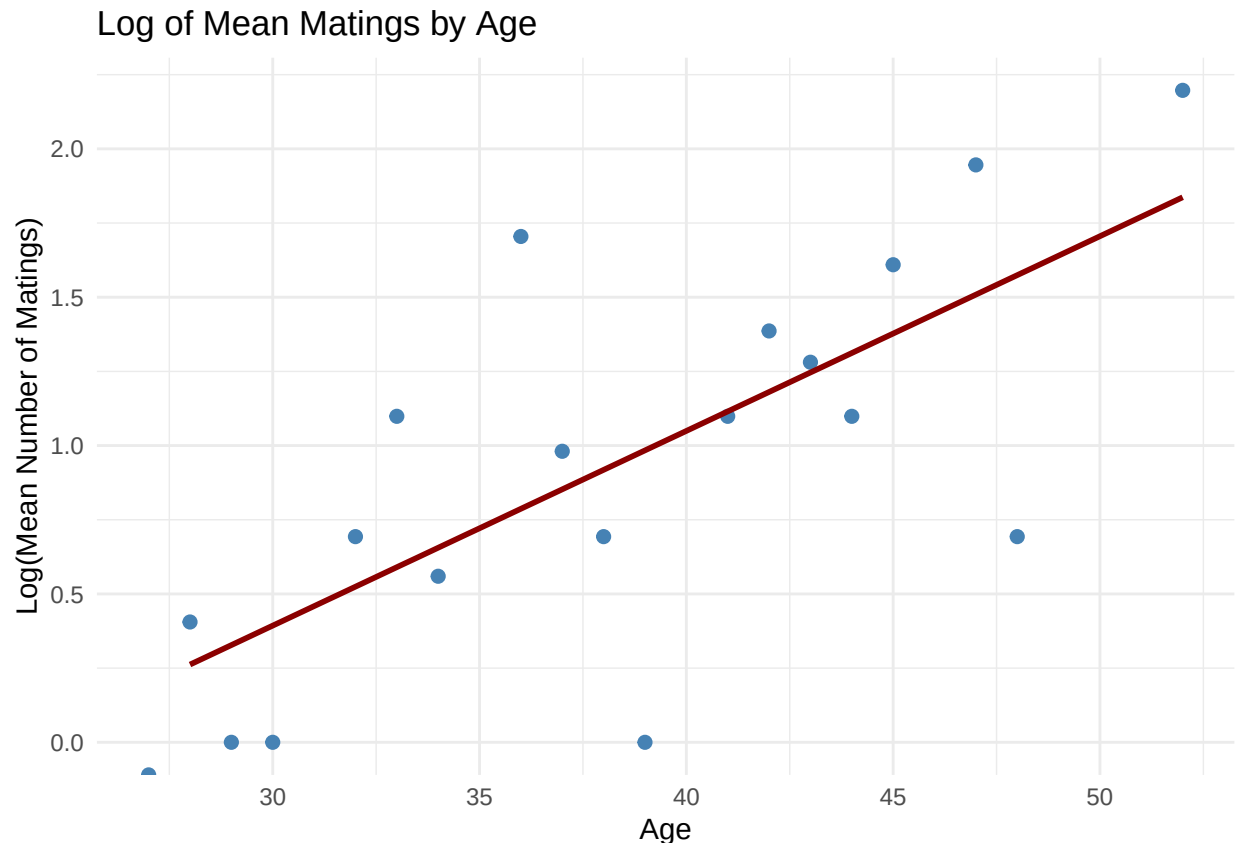
(c) For each age, calculate the mean number of matings. Take the log of each mean and plot it by AGE.

```
library(dplyr)

# Group by AGE and calculate the mean number of matings
age_means <- data %>%
  group_by(AGE) %>%
  summarize(mean_matings = mean(MATINGS))
# Add a column with log of mean matings
age_means <- age_means %>%
  mutate(log_mean_matings = log(mean_matings))
# Scatterplot of log(mean matings) by AGE
ggplot(age_means, aes(x = AGE, y = log_mean_matings)) +
  geom_point(color = "steelblue", size = 2) +
  geom_smooth(method = "lm", se = FALSE, color = "darkred") +
  labs(
    title = "Log of Mean Matings by Age",
    x      = "Age",
    y      = "Log(Mean Number of Matings)"
  ) +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
## ('stat_smooth()').
```



(i) This plot helps us check if the log of mean matings increases linearly with age. That's important because a Poisson regression assumes a linear relationship between the predictor (AGE) and the log of the expected count (mean matings).

(ii) Not really. The plot shows a mostly increasing pattern without a clear peak or drop afterward. For a quadratic trend, we'd expect the points to rise and then fall (or vice versa), which isn't happening here. So there's no strong evidence of a quadratic relationship.

(d) Fit a Poisson regression model with a linear term for AGE. Exponentiate and then interpret the coefficient for AGE.

```
# Fit the Poisson regression model
model_d <- glm(MATINGS ~ AGE, data = data, family = poisson)
summary(model_d)
```

```
##
## Call:
## glm(formula = MATINGS ~ AGE, family = poisson, data = data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.58201    0.54462  -2.905  0.00368 **
## AGE          0.06869    0.01375   4.997 5.81e-07 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 75.372  on 40  degrees of freedom
## Residual deviance: 51.012  on 39  degrees of freedom
## AIC: 156.46
##
## Number of Fisher Scoring iterations: 5
```

```
# Exponentiate the coefficient for AGE
exp(coef(model_d)["AGE"])
```

```
##      AGE
## 1.071107
```

- We fit a Poisson regression with AGE as the predictor. The estimate for AGE is 0.06869, and it's statistically significant ($p < 0.001$), so age has a real impact on the number of matings.
- When we exponentiate the coefficient, we get 1.0711. That means for every one-year increase in age, the expected number of matings increases by about 7.1%, assuming everything else stays the same.

(e) Construct a 95% confidence interval for the slope and interpret in context. (you may want to exponentiate endpoints).

```
# Construct 95% confidence interval for AGE coefficient
confint(model_d)["AGE", ]
```

```
## Waiting for profiling to be done...
```

```
##      2.5 %      97.5 %
## 0.04167776 0.09563762
```

```
# Exponentiate the interval to interpret in terms of rate ratio
exp(confint(model_d)["AGE", ])
```

```
## Waiting for profiling to be done...
```

```
##      2.5 %      97.5 %
## 1.042558 1.100360
```

- We're 95% confident that for every one year increase in age, the expected number of matings increases by between 4.26% and 10.04%. Since the entire confidence interval is above 1, this supports the idea that older elephants tend to have higher mating rates.

(f) Are the number of matings significantly related to age? Test with wald test and drop-in deviance test

```
# (i) Wald test - check summary output for AGE p-value
summary(model_d)
```

```
##
## Call:
## glm(formula = MATINGS ~ AGE, family = poisson, data = data)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.58201    0.54462  -2.905  0.00368 **
## AGE          0.06869    0.01375   4.997 5.81e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
## Null deviance: 75.372  on 40  degrees of freedom
## Residual deviance: 51.012  on 39  degrees of freedom
## AIC: 156.46
##
## Number of Fisher Scoring iterations: 5
```

```
# (ii) Drop-in-deviance test
model_null <- glm(MATINGS ~ 1, data = data, family = poisson)
anova(model_null, model_d, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: MATINGS ~ 1
## Model 2: MATINGS ~ AGE
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         40      75.372
## 2         39      51.012  1    24.36 7.991e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- (i) Wald test: From the model summary, the p-value for AGE is 0.000000581 which is far below 0.05. This means AGE is a highly significant predictor of the number of matings.
- (ii) Drop-in-deviance test: Comparing the null model to the model with AGE gives a p-value of 0.0000007991, also far below 0.05. Adding AGE to the model significantly improves the fit.

(g) Add a quadratic term in AGE to determine whether there is a maximum age for the number of matings for elephants. Is a quadratic model preferred to a linear model? To investigate this question, use the wald test and the drop in deviance test.

```

# Create quadratic term
data$AGE2 <- data$AGE^2
# Quadratic model
model_quad <- glm(MATINGS ~ AGE + AGE2, data = data, family = poisson)
# (i) Wald test - p-value for AGE2
summary(model_quad)

```

```

##
## Call:
## glm(formula = MATINGS ~ AGE + AGE2, family = poisson, data = data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -2.8574060  3.0356383  -0.941   0.347
## AGE          0.1359544  0.1580095   0.860   0.390
## AGE2        -0.0008595  0.0020124  -0.427   0.669
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 75.372  on 40  degrees of freedom
## Residual deviance: 50.826  on 38  degrees of freedom
## AIC: 158.27
##
## Number of Fisher Scoring iterations: 5

```

```

# (ii) Drop-in-deviance test
anova(model_d, model_quad, test = "Chisq")

```

```

## Analysis of Deviance Table
##
## Model 1: MATINGS ~ AGE
## Model 2: MATINGS ~ AGE + AGE2
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         39      51.012
## 2         38      50.826  1   0.18544   0.6667

```

- (i) Wald test: The p-value for AGE Squared is 0.669 which is greater than 0.05, meaning there is no statistically significant evidence that a quadratic term improves the model.
- (ii) Drop-in-deviance test: Comparing the linear and quadratic models gives a p-value of 0.667 which is greater than 0.05, so adding AGE Squared does not significantly improve model fit.

(h) What can we say about the goodness-of-fit of the model with age as the sole predictor? Compare the residual deviance for the linear model to a χ^2 distribution with the residual model degrees of freedom.

```
# (i) Goodness-of-fit test
pchisq(deviance(model_d), df.residual(model_d), lower.tail = FALSE)
```

```
## [1] 0.09426231
```

```
# (ii) Fit quasi-Poisson model
model_quasi <- glm(MATINGS ~ AGE, data = data, family = quasipoisson)
# Compare summaries
# Poisson model
summary(model_d)
```

```
##
## Call:
## glm(formula = MATINGS ~ AGE, family = poisson, data = data)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.58201    0.54462  -2.905  0.00368 **
## AGE          0.06869    0.01375   4.997 5.81e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 75.372  on 40  degrees of freedom
## Residual deviance: 51.012  on 39  degrees of freedom
## AIC: 156.46
##
## Number of Fisher Scoring iterations: 5
```

```
# Quasi-Poisson model
summary(model_quasi)
```

```
##
## Call:
## glm(formula = MATINGS ~ AGE, family = quasipoisson, data = data)
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.58201    0.58590  -2.700  0.0102 *
## AGE          0.06869    0.01479   4.645 3.81e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasipoisson family taken to be 1.157334)
##
##      Null deviance: 75.372  on 40  degrees of freedom
## Residual deviance: 51.012  on 39  degrees of freedom
## AIC: NA
##
## Number of Fisher Scoring iterations: 5
```



```
# (iii) Estimated dispersion parameter
dispersion_param <- deviance(model_d) / df.residual(model_d)
dispersion_param
```

```
## [1] 1.30799
```

- (i) The estimated coefficients for both intercept (-1.58201) and AGE (0.06869) are identical to the Poisson model. This is expected because changing to quasi-Poisson doesn't change the point estimates, only the estimated standard errors
- (ii) The standard errors increase slightly: Intercept SE changes from 0.54462 to 0.58590 and the AGE SE changes from 0.01375 to 0.01479. This widening basically reflects adjustment for over dispersion.
- (iii) We get 1.308 from the dispersion_param. Since this is greater than 1, there is over dispersion.
- (iv) Since the estimated dispersion parameter (1.308) is greater than 1, this suggests over dispersion. When we adjust for over dispersion, the standard errors get larger. Larger standard errors mean smaller test statistics and bigger p-values. As a result, we are less likely to find a coefficient significant after adjusting, because the model is accounting for extra variability that the basic Poisson model ignored.